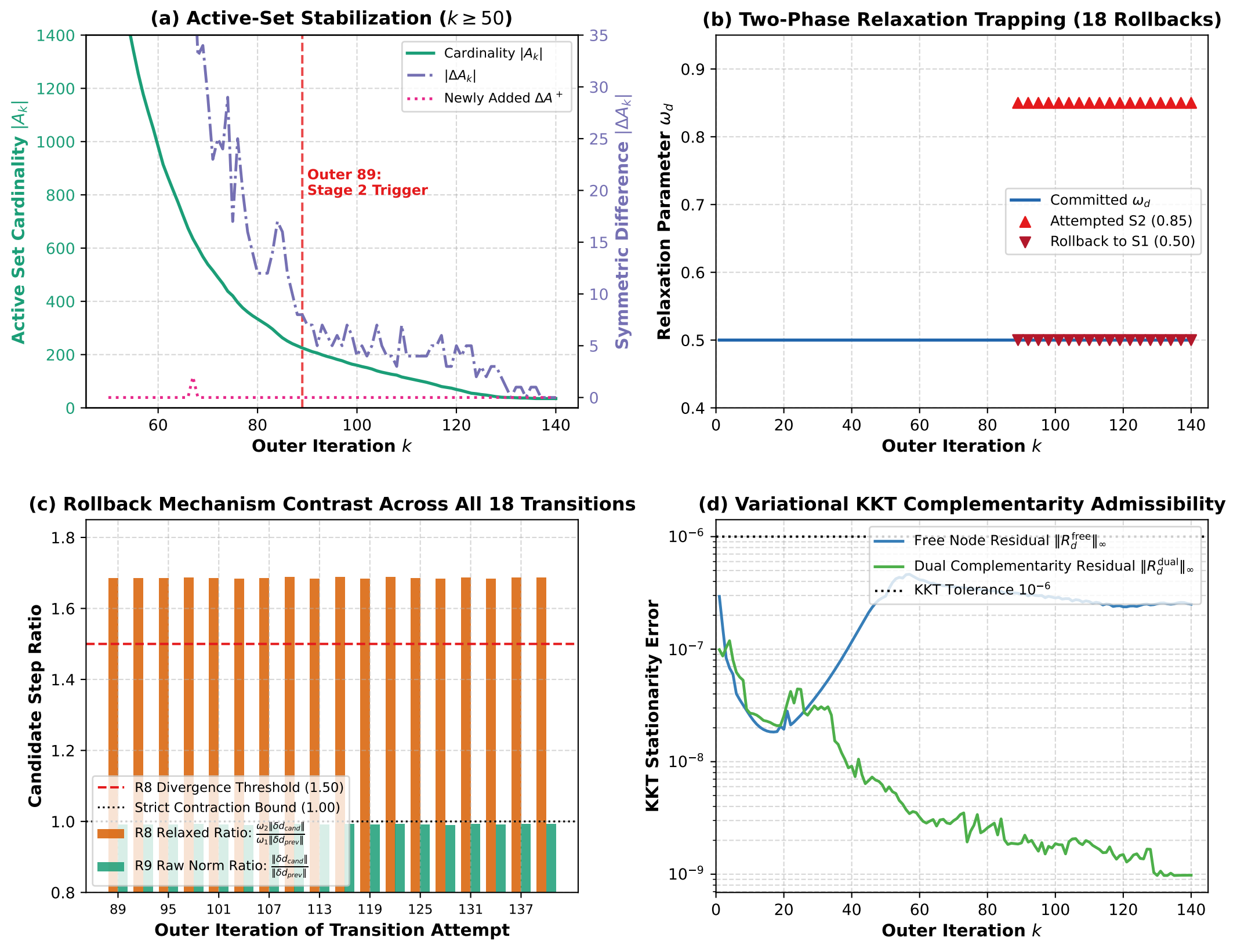
(a) Active-Set Stabilization ($k \geq 50$)
Active Set Cardinality $|A_k|$
Symmetric Difference $|\Delta A_k|$
Outer Iteration k
Cardinality $|A_k|$
$|\Delta A_k|$
Newly Added ΔA^+
Outer 89:
Stage 2 Trigger
(b) Two-Phase Relaxation Trapping (18 Rollbacks)
Relaxation Parameter ω_d
Outer Iteration k
Committed ω_d
Attempted S2 (0.85)
Rollback to S1 (0.50)
(c) Rollback Mechanism Contrast Across All 18 Transitions
Candidate Step Ratio
Outer Iteration of Transition Attempt
R8 Divergence Threshold (1.50)
Strict Contraction Bound (1.00)
R8 Relaxed Ratio: $\frac{\omega_2\|\delta d_{cand}\|}{\omega_1\|\delta d_{prev}\|}$
R9 Raw Norm Ratio: $\frac{\|\delta d_{cand}\|}{\|\delta d_{prev}\|}$
(d) Variational KKT Complementarity Admissibility
KKT Stationarity Error
Outer Iteration k
Free Node Residual $\|R_d^{\text{free}}\|_\infty$
Dual Complementarity Residual $\|R_d^{\text{dual}}\|_\infty$
KKT Tolerance 10^{-6}